EART11300 Semester 2 Microscope Practical Class 2
Igneous, sedimentary and metamorphic rocks under the microscope

In this practical you will

- practise describing optical properties of minerals and begin to identify a few important rock-forming minerals under the microscope,
- see how the textures of rocks at thin-section scale can be used to classify them as igneous, sedimentary or metamorphic,
- practise drawing sketches of thin sections.

There are 9 thin sections to look at. Your tasks are to (1) answer the questions about the optical properties of specific minerals in each sample, (2) draw sketches as suggested*, and (3) use the attributes of rocks in thin section (on page 5, essentially the same as you used for hand specimens last semester) to determine the rock type. Task 3 has been done for you for the first 3 samples. Make sure you agree with the answers for these 3 samples before moving on to the unknown samples.

* When you draw sketches today, you should add a scale bar. To do this, you will need to measure the width of the field of view. We will do this together at the start of the practical.

Peridotite (RP or R56)

This is an igneous rock. Most of the Earth's mantle is made of peridotite, but these samples actually comes from a large intrusion (from the island of Rum, Inner Hebrides, Scotland). You looked at R56 last week; RP is similar. The main mineral is **olivine**. It displays quite respectable crystal shapes and well-developed irregular fractures. Draw a few crystals that show a reasonable crystal shape (they will have parallel sides). Is its relief high, or low?
Can you see any cleavage in it? Note the range of interference colours represented by the various crystal orientations. What is the **maximum** colour (on the Michel Levy colour chart) that you can see?

Now look at the evidence that this is an igneous rock, using the possible attributes of igneous rocks:

- Does it have interlocking crystals? No.
- Do some crystals have their own shape? Yes (not so obvious in R56).
- Does it contain relatively few minerals? Yes, most of the rock is made of olivine plus two other minerals (there are also a few opaque grains and some alteration).
- Are crystals of each mineral quite uniform? Yes.
- Does it contain fossil gas bubbles (= vesicles)? No, it has no porosity.
- Do the crystals have a preferred orientation? No.

Blueschist (A/986)

This is a metamorphic rock. Before metamorphism it was an igneous rock (probably volcanic), and it has been metamorphosed at high pressure, producing several new minerals. The most distinctive new mineral is the blue one (which you do **not** need to be able to identify). Look at the **pleochroism** of this mineral. What pleochroic colours can you see?
There is a mineral in this thin section that is equant (see 2-D sections through 3-D crystals on page 6 of last week's practical) and isotropic. Look for it and draw a sketch of a crystal. What is its relief? What is its colour? This mineral is garnet.

Now look at the evidence that this is a metamorphic rock, using the possible attributes of metamorphic rocks:

- Does it have compositional layering? Yes. Look at the slide with the naked eye to see this.
- Does it contain relatively few minerals? Yes, 5 or 6.
- Are crystals of each mineral quite uniform? Yes.
- Do some crystals have their own shape? Yes.
- Does it contain any porosity? No.
- Do the crystals have a preferred orientation? Yes, it has foliation. This is the key evidence that this is a metamorphic rock.

Sandstone (R25 or SK)

This is a sedimentary rock. It contains 2 minerals, one with low interference colour, one with high interference colour. Look at the low interference colour mineral. Does it contain any cleavage? This mineral is **quartz**. Notice how the crystals are all very clear – there is no alteration. This is a useful property of quartz. Other properties of quartz are: low relief, no twinning or zoning (which you will see in some low-interference colour minerals in other thin sections).

The other mineral, with high interference colour, is **calcite**. The sandy interference colour, with green and pink tinges, is a typical high interference colour. Calcite (and its fellow carbonate mineral, dolomite) is the only common mineral you will see with high interference colour. Notice how the calcite is filling in the gaps around the quartz crystals, forming a cement. Sketch a few quartz grains and annotate your sketch to show the calcite cement.

Now look at the evidence that this is a sedimentary rock, using the possible attributes of sedimentary rocks:

- Does it consist of particles (clasts) of pre-existing rocks? Yes, many of the quartz grains have a very irregular shape, indicating that they are not fresh crystals but have been abraded from a pre-existing rock.
- Are the clasts rounded? Some of them.
- Does it contain fossils? No.
- Does it contain organic matter? No.
- Does it contain any porosity? No (the calcite cement holding the quartz grains together has filled any earlier porosity).
- Is it layered? Not that you can see on the thin section scale.

Rocks for you to identify:

SB2

Before working out what rock type this is, look at the green-brown mineral. It has very good cleavage – make sure you can see this. Draw a crystal showing good cleavage. Now rotate the stage so that the cleavage in this crystal is oriented N-S. Which of the following pleochroic colours can you see: pale brown or dark green-brown? Now rotate the stage so that the cleavage in this crystal is oriented E-W. Which of the following pleochroic colours can you see: pale brown or dark green-brown? This mineral is **biotite** (usually more of an orange-brown than a green-brown). Notice that is full of tiny inclusions with pleochroic haloes. These inclusions are the mineral zircon, which contain small amounts of radioactive uranium that emit alpha radiation that causes damage to the biotite. These are quite common in biotite.

Use the attributes of rocks in thin section on page 5 to determine what rock type this is, giving reasons for your answer.

NH

Before working out what rock type this is, look at the crystals of the colourless mineral in between the green crystals (that you looked at last week). Is their interference colour low, intermediate or high? Notice how they are twinned (see description below). This mineral is **feldspar** (plagioclase feldspar, to be precise, but we are just going to call it feldspar today).

Use the attributes of rocks in thin section on page 5 to determine what rock type this is, and draw a sketch of a few crystals to illustrate this.

TWINNING. Some minerals tend to develop crystals composed of two or more distinct parts, chemically identical, but of different orientation. Some twinned crystals consist of two parts **(simple twinning),** while others comprise many parallel 'twin' units within a single crystal **(multiple twinning).** In thin section, twinning is most effectively seen under crossed polars, because the individual units tend to display different interference colours and/or extinction positions, and yet can be clearly seen to belong to a single crystal in terms of the other optical properties. The boundaries between the units are typically plane surfaces, and appear as straight lines in thin section, provided that they are approximately perpendicular to the surface of the section.

CP

This rock contains 3 main minerals plus an opaque phase and some alteration minerals.
These 3 minerals are:

1. A pale brown mineral. Is this mineral pleochroic? Does it have cleavage?
   Is its interference colour low, intermediate or high?
2. A colourless mineral with high relief. Does this mineral have cleavage? Fractures?
   Is its interference colour low, intermediate or high?
   Notice the alteration in the fractures – this is common for this mineral, which is **olivine**.
3. A colourless mineral with low relief. Is its interference colour low, intermediate or high?
   Notice how this mineral is twinned (see description above). It is **feldspar** (plagioclase feldspar, to be precise, but we are just going to call it feldspar today).

Use the attributes of rocks in thin section on page 5 to determine what rock type this is, and draw a sketch of a few crystals to illustrate this.

CG

The main minerals in this rock are:

1. **quartz** – see the R25/SK description
2. **feldspar** – see the NH and CP descriptions. In this rock the feldspar is heavily altered (to flecks that are either grey under crossed polars or have bright interference colours). This is common in feldspar, whereas quartz is never altered.
3. **mica**. Most of the mica is **biotite**. You have already seen biotite in SB2, but the biotite in this rock is a more typical red-brown colour. As for SB2, draw a biotite crystal showing good cleavage, and describe the pleochroic colour when the cleavage is oriented N-S, and when it is oriented E-W. Note that you also looked at biotite last week in 15216.

Use the attributes of rocks in thin section on page 5 to determine what rock type this is, and explain your answer.

S10

The most abundant mineral in this rock occurs in a wide range of grain sizes. It is the same mineral as in at least one of the samples above. Can you identify it from its optical properties?

Use the attributes of rocks in thin section on page 5 to determine what rock type this is, and draw a sketch to illustrate your answer.

RE114

This sample contains crystals of one mineral that are up to ~5 mm long. The rest of the rock is very fine-grained. The large crystals are the same mineral as in at least one of the samples above. Can you use its optical properties to identify it?

Use the attributes of rocks in thin section on page 5 to determine what rock type this is, and draw a sketch to illustrate this.

Attributes of the three categories of rock in hand specimen or thin section:

Igneous Rocks

These rocks have crystallised from partially molten material – magma – either within the Earth or on Earth's surface.

When you are looking at a hand specimen or thin section:

1. Igneous rocks tend to have interlocking crystals
2. Some crystals have their own shape (i.e. are bounded by flat crystal faces) indicating unhindered growth from a liquid.
3. Contain relatively few minerals (normally 5 different mineral species or less).
4. Crystals of each mineral tend to be quite uniform.
5. May contain fossil gas bubbles (= vesicles). Not otherwise porous.
6. Crystals don't usually have a preferred orientation (but there are exceptions, e.g. cumulates and rocks with flow-aligned crystals)
7. Do not contain organic matter or fossils.

Sedimentary rocks

These rocks have formed by the accumulation, as layered deposits on the Earth's surface, of material derived by weathering of pre-existing rocks or material of biological origin.

When you are looking at a hand specimen or thin section:

1. Commonly consist of particles (clasts) of pre-existing rocks, which may include polycrystalline rock fragments.
2. Clasts may be rounded, indicating abrasion during sediment transport.
3. May contain fossils.
4. May contain organic matter.
5. May be quite porous.
6. Commonly layered (bedded), often with dramatic compositional contrasts. (May only see this when looking at a large scale, e.g. on maps or in the field.)

Metamorphic rocks

These rocks have formed by alteration of pre-existing igneous or sedimentary rocks, <u>in the solid state</u>, through the action of heat and/or pressure.

When you are looking at a hand specimen or thin section:

1. Compositional layering (fossil bedding) may be present, but may be deformed (folded, etc).
2. Contain relatively few minerals (normally 6 or less); crystals of each mineral tend to be quite uniform.
3. Crystals often have a preferred orientation (tectonic fabric) produced by deformation. If the fabric is planar (foliation), it may be discordant to any layering present.
4. Some crystals have their own shape; others have shapes indicative of competitive growth in the solid state (analogy: bubbles in foam).
5. No significant porosity.
6. No fossils usually survive.